

Weighted +I correction: paired exact-pattern simulation

Same multinomial alignment counts evaluated by both formulas; one-sided alpha = 0.05

Legacy counterfactual

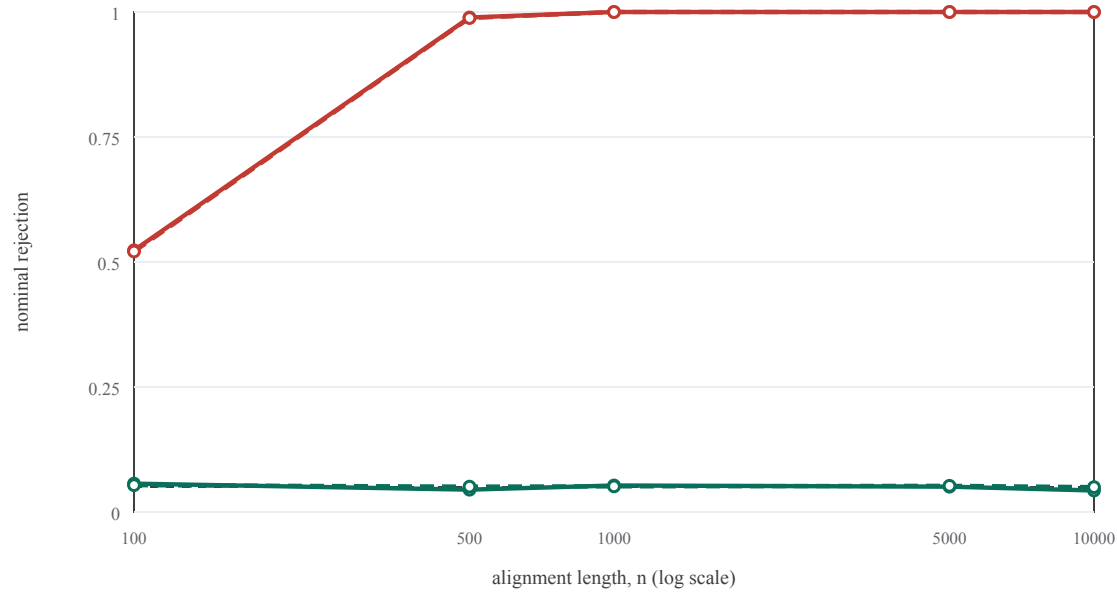
Refined invariant null

+I+G4 (solid in A; blue in D)

+I+R4 (dashed in A; pink in D)

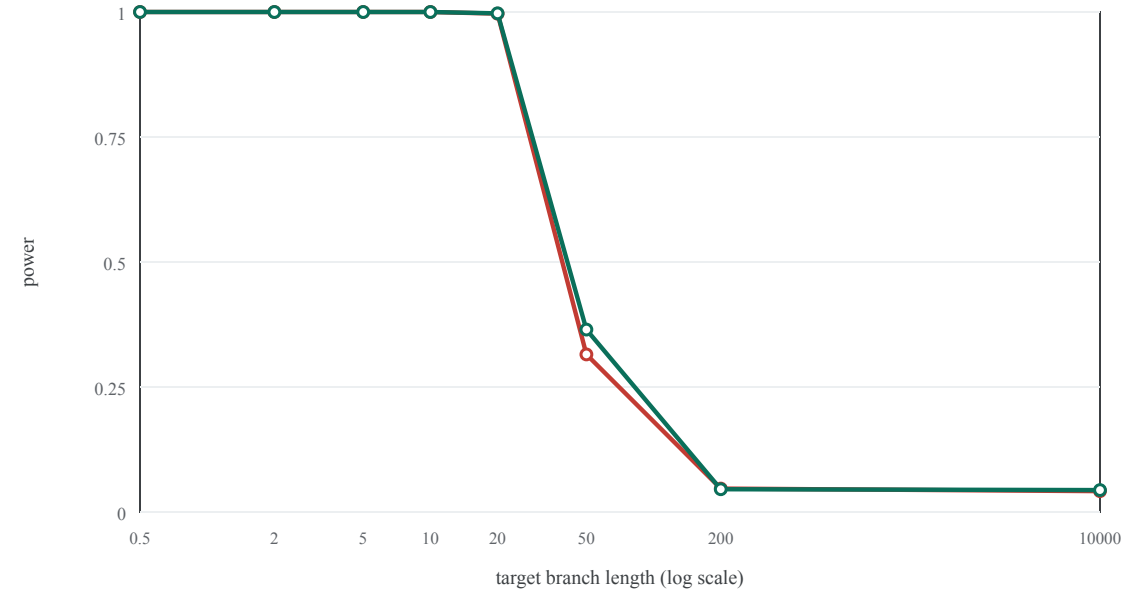
A Extreme-branch false positives

t = 10,000 under the correct +I null



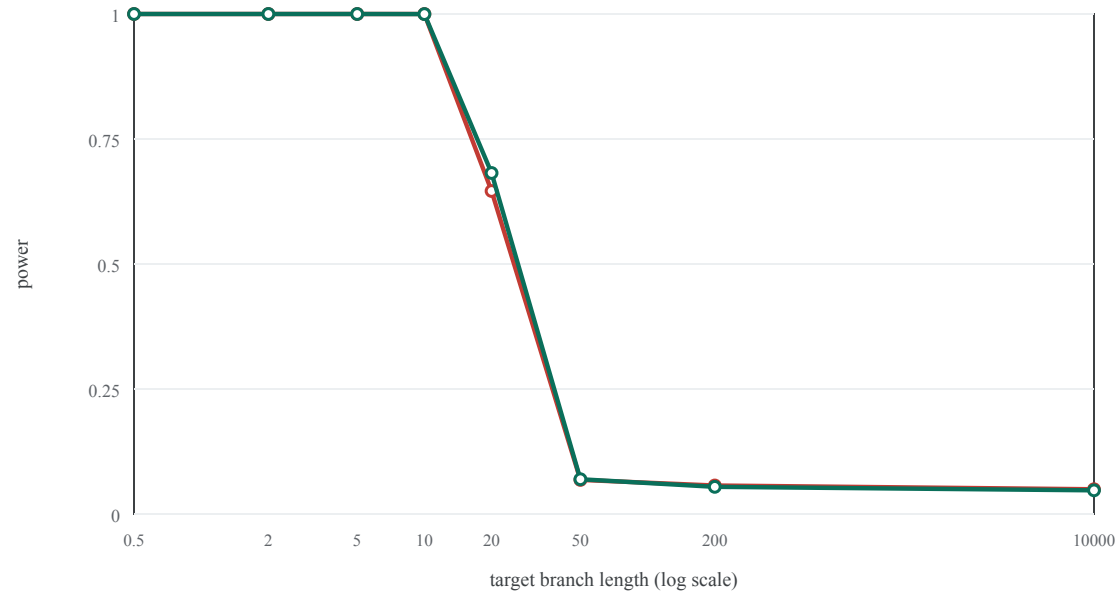
B +I+G4 calibrated power

n = 1,000; formula-specific training-null threshold



C +I+R4 calibrated power

n = 1,000; formula-specific training-null threshold



D Paired calibrated-power difference

refined minus legacy; t = 20

